

Shekh Humaira Sunzida

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Education

University of Dhaka, Bangladesh

September, 2025 - Present

Master of Science (Thesis Based) in Computer Science & Engineering (CSE)

Expected Graduation Date: March, 2027

Comilla University, Bangladesh

2019 - 2024

Bachelor of Science (Engineering) in Information & Communication Technology (ICT)

GPA: 3.50 (out of 4.00), Class: 1st class, Graduation Date: 27 March, 2024

Research Publications

Pitch Extraction Using DCT Based Power Spectrum Method in Noisy Speech

September, 2025

Humaira Sunzida and et al., DOI: [10.11591/ijaas.v14.i3.pp955-965](https://doi.org/10.11591/ijaas.v14.i3.pp955-965)

International Journal of Advances in Applied Sciences (IJAAS) Vol. 14, No. 3, September 2025

Research Experience

Parents-Daughter Communication regarding RH and Puberty Health:

A Study among female students in Bangladesh (Paid)

April, 2024 - July, 2024

Supervised By: Amena Begum, Assistant Professor, Comilla University

- A study of 1,200 female students across eight regions of Bangladesh was conducted & response processed using python.
- A CSV file was created using python & fed to SPSS for further analysis & graphical representation which enrich my capability to bridge between statistical tool with machine learning.

Pitch Extraction Using DCT Based Power Spectrum Method in Noisy

Speech (BSc. Research Project)

January, 2023 - January, 2024

Supervised By: Dr. Md. Saifur Rahman, Associate Professor, Comilla University

- A research conduct to develop a efficient method of extracting Fundamental Frequency of human voice using Discrete Cosine Transform (DCT) which enhanced my ability to apply scientific & mathematical theory to quantitative data.
- Computer based research project, includes MATLAB for coding, simulation & visual representation of data as well as excel for conducting performance & accuracy measurement.
- Each investigation involves a written report in LATEX along with their visual representation.

Fake Currency Detection

December, 2022 - May, 2023

Supervised By: Alimul Rajee, Lecturer, Comilla University

- A project developed using python, analyzes key features of the currency that improves my data manipulation capabilities with python.
- Features extracted leveraging data preprocessing & image processing techniques to distinguish between genuine & counterfeit notes.

Skills & Proficiency

Software Proficiency :

Matlab, SPSS, Jupyter Notebook, Xampp, MySQL, SQLite, Android Studio, Git, Scilab.

Language Proficiency :

Python, C, C++, R, JAVA, SQL, HTML, CSS, PHP, Javascript, Typescript, LaTeX.

Familiar with:

Matplotlib, Seaborn, Speech Recognition, Pandas, NumPy, Selenium, Scikit-learn.

Research Interest

Machine Learning, Artificial Intelligence & Data Science.

Academic Projects

AI Virtual Assistant

- **Artificial Intelligence** based project developed using **Python** that Assist human in daily activities according to command.
- Capable of custom conversations, performing Wikipedia/Google/YouTube searches, sending emails, predicting date/time/weather, telling jokes/fun facts, delivering latest news & even capturing images to create sketches etc. providing versatile & engaging user experience.
- Implemented my knowledge earned from “Google IT Automation with Python” by COURSERA & other resources into this real world project.

FingerPrint Detection

- A python based project that securely authenticates identity by analyzing unique fingerprint features.

Arduino Voice Controlled Bluetooth Car

- An IoT based project that Automates vehicle using voice command through mobile app.
- Applied my “Aduino Programming & Hardware Fundamentals By Hackstar” course learning to gather practical experience on how to work with microprocessor & microcontroller.

Movie Recommendation System

- Collaborative-based recommendation system using NearestNeighbors to suggest similar movies based on user-selected titles.
- Evaluate the output difference for Cosine & Euclidean distance.

Phone Book App

- An App development based project using Android Studio & Java that allows users to efficiently store, manage and search contacts.

Achievements

- Achieved a National Science and Technology (NST) Fellowship 2024-25 on “EnviroBot (A Mobile Robotic Arm for Efficient Cleaning and Task Automation)”.
- Champion in Project Showcase on “Smart CoU” in 2ND COUSC NATIONAL SCIENCE FESTIVAL-2024
- Champion IN INNOVATION SHOWCASING-2024 hosted by University Grants Commission (UGC) on “GPS location System”.

Professional Experience

Associate SQA Engineer, Orbitax Bangladesh Limited

September,2025 - Present

- Ensure quality and reliability of the products that unifies international tax processes.
- Performed functional, regression & exploratory testing while designing and executing test cases.
- Write comprehensive automation script with Jmeter to conduct API testing & performance testing.

Teaching Experience

- Co-Founder & Mentor of RoboICT (A nonprofitable organization provide practical tech learning with lab facilities to students)
- Earned an excellent reputation for conducting personalized as well as group tutoring sessions on Mathematics, Information & Communication technology(ICT).

Certifications

- Google Data Analytics Certification by COURSERA.
- Supervised Machine Learning & Advanced Learning Algorithm Certification by COURSERA.
- Google IT Automation with Python Certification by COURSERA.
- Google IT Support Certification by COURSERA.

Standardized Tests

GRE : Quantitative Reasoning : 155, Verbal Reasoning : 148, Analytical Writing Assessment : 2.5
TOEFL : Reading : 20, Listening : 27, Speaking : 23, Writing : 22

Co-Curricular Activites

- Achieved a **4-Year Degree** on Singing from Chhayanaut Shongshkriti-Bhobon
- Performed live music at various events, showcasing creativity, stage presence & collaboration skills.